

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.0003446247514520339
rotor delay [s]: 0.199999997433
gimbal delay [s]: 0.030751303016
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 5.3444260254306806e-17
common-scale FD null norm: 8.274807140417531e-12

Dimensionless coordinate

log_mass_scale	-0.260085894189
log_second_moment_mode_xx	0.82314108788
log_second_moment_mode_yy	1.09989046035
log_second_moment_mode_zz	0.0861898900633
log_second_moment_mode_xy	-0.0683342859697
log_second_moment_mode_xz	0.344391320648
log_second_moment_mode_yz	1.25521829327
dimensionless_cog_displacement_x	-0.190750263879
dimensionless_cog_displacement_y	0.0172682863631
dimensionless_cog_displacement_z	-0.222819686416
log_force_effectiveness_1	-0.282168276354
log_force_effectiveness_2	-0.587497283509
log_force_effectiveness_3	-0.563477365612
log_force_effectiveness_4	-0.315992618632

Physical representative

mass [kg]: 1.8130473146511679
CoG body [m]: [-0.0385696 0.00327782 -0.03317916]
force effectiveness: [0.75414677 0.55571635 0.56922621 0.72906483]
inertia [kg m^2]:
[0.2540674256, -0.005679984979, -0.002226702725]
[-0.005679984979, 0.1511186582, -0.01059593931]
[-0.002226702725, -0.01059593931, 0.4034626597]
principal moments [kg m^2]: [0.15035365 0.25436192 0.40393317]

Per-bag raw residual-wrench RMS

failure_2_single: [0.14068909 0.23370896 0.33396316 0.02607659 0.07521601 0.042437]